Last time:

(Y, d_Y) is a subspace of (X, d_X) $\iff \begin{cases} \cdot\ \emptyset \neq Y \subseteq X \\ \cdot\ d_Y(x,y) = d_X(x,y) \quad \forall x,y \in Y \end{cases}$

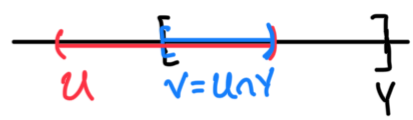
Prop Let (X, d_X) be a metric space and (Y, d_Y) a subspace (with the induced metric).

① A set $V \subseteq Y$ is open in (Y, d_Y) $\iff$ $\exists\, \mathcal{U} \subseteq X$ open in (X, d_X) s.t. $V = \mathcal{U} \cap Y$.

② A set $G \subseteq Y$ is closed in (Y, d_Y) $\iff$ $\exists\, F \subseteq X$ closed in (X, d_X) s.t. $G = F \cap Y$.

Ex ① $[0,1)$ is not open in $(\mathbb{R}, |\cdot|)$.

$[0,1)$ is open in $([0,2], |\cdot|)$, since $[0,1) = (-1,1) \cap [0,2]$.



② $[0,1)$ is not closed in $(\mathbb{R}, |\cdot|)$.

$[0,1)$ is closed in $((-1,1), |\cdot|)$, since $[0,1) = (-1,1) \cap [0,2]$.

Pf: ① $\Rightarrow$: Suppose V is open in (Y, d_Y). Then $\forall a \in V$, $\exists\, r_a > 0$ s.t. $B^Y_{r_a}(a) \subseteq V$. Note that

$$B^Y_{r_a}(a) = \{ y \in \underline{Y} : \underline{d_Y(a,y)} < r_a \}$$

$$\subseteq \qquad\qquad =$$

$$B^X_{r_a}(a) = \{ x \in \underline{X} : \underline{d_X(a,y)} < r_a \}$$

So $B^Y_{r_a}(a) = B^X_{r_a}(a) \cap Y$, and

$$V = \bigcup_{a \in V} B^Y_{r_a}(a) = \bigcup_{a \in V} [B^X_{r_a}(a) \cap Y]$$

$$= \underbrace{\left[\bigcup_{a \in V} B^X_{r_a}(a)\right]}_{\text{Set } = \mathcal{U}} \cap Y$$

Then $\mathcal{U}$ is open in (X, d_X), since the arbitrary union of open sets is open.

$\underline{\Leftarrow 1}$: Suppose $V = \mathcal{U} \cap Y$ for some $\mathcal{U}$ open in (X, d_X). Want: V is open in (Y, d_Y). Fix $a \in V$. Then $a \in \mathcal{U}$, so $\exists r > 0$ s.t. $B^X_r(a) \subseteq \mathcal{U}$. So

$$B^Y_r(a) = B^X_r(a) \cap Y \subseteq \mathcal{U} \cap Y = V.$$

② $G \subseteq Y$ is closed in (Y, d_Y)

$\iff Y \setminus G$ is open in (Y, d_Y)

$\overset{①}{\iff} \exists\, \mathcal{U}$ open in (X, d_X) s.t. $Y \setminus G = \mathcal{U} \cap Y$.

Note that

$$G = Y \setminus (Y \setminus G) = Y \setminus (\mathcal{U} \cap Y) = Y \cap (\mathcal{U} \cap Y)^c$$

$$= Y \cap (\mathcal{U}^c \cup Y^c) = (Y \cap \mathcal{U}^c) \cup \underbrace{(Y \cap Y^c)}_{= \emptyset} = Y \cap \mathcal{U}^c$$

So set $F = \mathcal{U}^c$. Moreover, $\mathcal{U}$ open in (X, d_X) $\iff F = \mathcal{U}^c$ is closed in (X, d_X). $\square$

<u>Cor</u> Let (X, d_X) be a metric space and (Y, d_Y) a subspace. The following are equivalent:

① Y is open (closed) in X.

② Any $A \subseteq Y$ that is open (closed) in Y is also open (closed) in X.

<u>Pf</u>: <u>② $\Rightarrow$ ①</u>: Take $A = Y$.

<u>① $\Rightarrow$ ②</u>: Suppose Y is open in X. Let $A \subseteq Y$ be open in Y. By the previous proposition, $\exists\, \mathcal{U}$ open in X s.t. $A = \underset{\substack{\uparrow \\ \text{open} \\ \text{in } X}}{\mathcal{U}} \cap \underset{\substack{\uparrow \\ \text{open} \\ \text{in } X}}{Y}$.

Then A is open in X, since the finite intersection of open sets is open. $\square$

<u>Prop</u> Let (X, d_X) be a metric space and (Y, d_Y) a subspace. For any $A \subseteq Y$, $\bar{A}^Y = \bar{A}^X \cap Y$.

<u>Pf</u>: For $a \in Y$,

$$a \in \bar{A}^Y \iff \forall r > 0, \quad \underbrace{B_r^Y(a)}_{= B_r^X(a) \cap Y} \cap A \neq \emptyset$$

$$\iff \forall r > 0, \quad B_r^X(a) \cap \underbrace{Y \cap A}_{= A} \neq \emptyset$$

$$\iff a \in \bar{A}^X \cap Y. \qquad \square$$

SEQUENCES

<u>Def</u> Let (X, d) be a metric space and let $\{x_n\}_{n \geq 1}$

be a sequence in X. We say $\{x_n\}_{n\geq 1}$ <u>converges</u> to $x \in X$ if: $\forall \varepsilon > 0 \ \exists N \in \mathbb{N}$ s.t.

$$d(x_n, x) < \varepsilon \quad \forall n \geq N.$$

When this is true, we write "$\lim_{n\to\infty} x_n = x$" or "$x_n \to x$ in (X, d) as $n \to \infty$".

<u>Ex</u> ① For $(\mathbb{R}, |\cdot|)$, we have $d(x_n, x) = |x_n - x|$. This agrees with the definition from Lecture 7.

② For $(\mathbb{R}^2, d_p)$, we have

$$d(x_n, x) = \left(|x_{n,1} - x_1|^p + |x_{n,2} - x_2|^p \right)^{\frac{1}{p}}.$$

<u>Rmk</u> $x_n \to x$ in (X, d) as $n \to \infty \iff$ the sequence of real numbers $d(x_n, x)$ converges to 0 as $n \to \infty$.

- Many of the facts that we proved for sequences in $\mathbb{R}$ are also true for metric spaces:

<u>Lem</u> The limit of a sequence is unique, if it exists.

<u>Thm</u> A sequence $\{x_n\}_{n\geq 1}$ converges to $x \iff$ every subsequence $\{x_{k_n}\}_{n\geq 1}$ converges to x.

- The proofs for $(\mathbb{R}, |\cdot|)$ also work for any metric space (X, d).

<u>Def</u> Let (X, d) be a metric space. We say that a sequence $\{x_n\}_{n\geq 1}$ is <u>Cauchy</u> if: $\forall \varepsilon > 0 \ \exists N \in \mathbb{N}$ s.t.

$$d(x_n, x_m) < \varepsilon \quad \forall\, n, m \geq N.$$

<u>Prop</u> If $\{x_n\}_{n\geq 1}$ converges, then it is Cauchy.

- Proof: Same as $(\mathbb{R}, |\cdot|)$ (see lecture 12)

<u>Rmk</u> Not every Cauchy sequence converges! E.g.,

① $((0,1), |\cdot|)$ is a metric space.

$\{\frac{1}{n}\}_{n\geq 1}$ is Cauchy, but does not converge in $(0,1)$.

② $(\mathbb{Q}, |\cdot|)$ is a metric space.

$$x_1 = 3, \qquad x_{n+1} = \frac{x_n}{2} + \frac{3}{2x_n} \quad \forall\, n \geq 1.$$

Then $\{x_n\}_{n\geq 1}$ is Cauchy, but doesn't converge in $\mathbb{Q}$.

- By lecture 12, this sequence converges to $\sqrt{3}$.